

S01 SOUND MAPPING | P

Intent : Incorporate strategic planning required to prevent issues of acoustic disturbance from various sources of noise.

Summary : This WELL feature requires projects to strategize an acoustical plan that identifies sources of noise that can negatively impact interior spaces.

Issue : Architectural trends, such as open workspaces, lightweight construction and exposed finishes or HVAC foster acoustically uncomfortable environments.¹⁻⁴ When noise from internal or external sources is increased in a space, occupants have been found to be easily distracted, less productive and susceptible to burnout.⁵⁻⁹ In office environments, employees care about privacy and collaboration.^{8,10} In one study from the United Kingdom, 99% of employees reported that their concentration was impaired by poor acoustical comfort in the workplace.⁷ Similar conditions are reported in workplaces worldwide.^{5,8,11-13} Some reports have shown that occupants are less likely to help others under high noise conditions, reducing collaboration in the workplace.⁶

Solutions : To address noise, floor plans should be designed with intent and use in mind.¹⁴ For example, a typical office can be categorized into four key types of programming: spaces for concentration, collaboration, socialization and learning.¹⁵ Location is important, since noise from social or collaborative spaces impacts spaces intended for concentration or learning.^{15,16} This approach can be implemented in any space type that incorporates spaces of socialization and recreation alongside areas for task-centric work or learning.¹⁵ These spaces can then be described as loud, quiet and mixed spaces to better assess the impact of sound on sensitive, quiet locations for concentration, learning or recreation.

PART 1 LABEL ACOUSTIC ZONES

For All Spaces:

The project provides the following:

- a. A floor plan or other design document showing the following acoustic zones throughout the project:
 1. Loud zone: includes areas intended for loud equipment or activities (e.g., mechanical rooms, AV/IT closets, kitchens, fitness rooms, social spaces, recreational rooms, music rooms).
 2. Quiet zone: includes areas intended for concentration, wellness, rest, study and/or privacy (e.g., restorative spaces, lactation rooms, nap rooms).
 3. Mixed zone: includes areas intended for learning, collaboration and/or presentation (e.g., auditoriums, classrooms, breakout spaces).
 4. Circulation zone: includes occupiable areas not intended for regular occupancy (e.g., hallways, egress, atria, stairs, lobbies)
 5. Not applicable zones: includes other areas without significant sources of sound (e.g., storage rooms, janitor rooms, coat closets) that are not regularly occupied.
- b. A plan for reprogramming or mitigating sound transmission between loud zones that border quiet zones (if any).

PART 2 PROVIDE ACOUSTIC DESIGN PLAN

For All Spaces:

The project provides one of the following:

- a. A plan developed by the project team and/or project owner that outlines acoustical solutions and a timeline for implementation with a focus on managing acoustical comfort, background noise, speech privacy, reverberation time and/or impact noise within the project boundary.
- b. A detailed report from a professional in acoustics that describes existing conditions, recommended solutions and measurement results with a focus on managing background noise, speech privacy, reverberation time and/or impact noise within the project boundary. These measurements are not required to adhere to the Performance Verification Guidebook requirements for on-site testing.

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S02 MAXIMUM NOISE LEVELS | O (MAX : 3 PT)

Intent : Achieve desired ambient noise levels such that HVAC, exterior noise intrusion or other noise sources do not impact occupant health and well-being.

Summary : This WELL feature prescribes maximum thresholds for ambient background noise that correspond to optimal levels of interior and exterior noise exposure.

Issue : All spaces have some degree of ambient background noise from HVAC equipment, exterior sources (e.g., traffic, outdoor equipment, pedestrians) or other building services. When the sum of these noise sources exceeds comfortable levels, the space may not function as intended. For instance, elevated levels of background noise can impact the perception of public address systems and diminish the perception of spoken word, which reduces critical listening ability and task performance.¹⁻⁶ Studies indicate that employees are unable to habituate to noise in office environments over time and office noise, with or without speech, can create stress and disrupt performance on more complex cognitive tasks (e.g., memory of prose, mental arithmetic).⁷⁻¹¹ Studies have shown that exposure to traffic noise can lead to increased risk of cardiovascular system issues, diabetes, hypertension, stroke, depression and high blood pressure.¹²⁻¹⁷ For children, chronic aircraft noise exposure impairs reading comprehension, mental arithmetic and proofreading.^{3,18-21}

Solutions : Interior noise sources can be controlled by selecting HVAC equipment with lower sound ratings and by designing the system to reduce sound within ducts.²² Exterior noise can be controlled by providing sound reduction at the building façade, windows and any exterior penetrations.²³ In both cases, these sound sources are easier to control when considered at the earliest possible stages of design.²⁴⁻²⁶

PART 1 LIMIT BACKGROUND NOISE LEVELS (MAX : 3 PT)

For All Spaces except Dwelling Units & Guest Rooms:

Note :

Category 1 Room Types:

- Areas for conferencing
- Areas for learning
- Areas for speaking

Category 2 Room Types:

- Enclosed areas for concentration

Category 3 Room Types:

- Open areas for concentration
- Areas with regularly used PA systems
- Areas for dining (excluding kitchenettes)

Category 4 Room Types:

- Areas with machinery and appliances used by occupants (e.g., baggage handling areas, security, commercial kitchens, kitchenettes, labs where spoken lectures do not take place)

The following requirement is met:

- a. Background noise levels do not exceed the following thresholds, as applicable:

	Tier Sound Pressure Level (SPL)	Category 4	Category 3	Category 2	Category 1 Points:
1	Average SPL (Leq)	dBA 55	50	45	40
		dBC 75	70	65	60
	90th percentile SPL (L10)	dBA 65	60	55	50
	Average SPL (Leq)	dBA 50	45	40	35
2		dBC 70	65	60	55
	90th percentile SPL (L10)	dBA 60	55	50	45

For Dwelling Units & Guest Rooms:

The following requirement is met:

- a. Average background noise levels in bedrooms, when measured over a 12-hour minimum time period (which must include the hours of 10 pm to 7 am), do not exceed 35 dBA (Leq).²⁷

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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S03 SOUND BARRIERS | O (MAX : 3 PT)

Intent : Increase the level of sound isolation and speech privacy between enclosed spaces.

Summary : This WELL feature requires that walls and doors meet a minimum degree of acoustical separation to provide adequate sound isolation and improve speech privacy.

Issue : Sound that transmits from one room to another through walls or doors can be distracting or annoying and also disturb sleep.¹⁻⁵ There is evidence that links noise annoyance in multi-story housing to poor mental health and perceived stress in residents.^{6,7} Speech privacy is also reduced when background sound in receiving rooms is lower as part of the room's intent (e.g., bedroom, restorative space, classroom).^{1,8} For walls, lightweight construction, glass and demountable partitions are typical in modern design and offer minimum acoustic separation and speech privacy, especially when wall construction is not uniform.⁹⁻¹¹

Solutions : Sound transmits through walls directly and around the construction by what is known as flanking. Walls with high sound transmission class ratings (STC) will provide sound isolation, only when the wall is constructed to reduce flanking at points where the wall connects to other building elements. Windows, glass fins or other penetrations diminish performance and should be used sparingly. Doors also reduce performance of walls but can be fitted with gaskets and seals to reduce sound transmission, when the door is closed. Rooms that require high speech privacy can use sound masking systems, in addition to high-performing walls to increase privacy.⁸

PART 1 DESIGN FOR SOUND ISOLATION AT WALLS AND DOORS (MAX : 1 PT)

For All Spaces:

The following requirements are met:

- Interior walls meet the following sound transmission class (STC) or weighted sound reduction (Rw) values. If an interior wall meets multiple categories listed, use the highest (i.e., more stringent) STC/Rw value listed.

Interior Wall Type	Minimum STC or Rw	OR	Minimum STC or Rw for Projects That Achieve Feature S06.1
Between loud zones and regularly occupied spaces.	60	OR	60
Between areas for conferencing, learning or sleeping and other regularly occupied spaces.	55	OR	50
Between adjacent quiet zones.	50	OR	45
Between rooms for concentration and other regularly occupied spaces.	45	OR	40
Between circulation zones and regularly occupied spaces.	40	OR	35

- Doors that connect two occupiable rooms and doors to mechanical equipment rooms have a non-hollow core, minimum STC/Rw of 30 and seals at the head, jamb and base.

PART 2 ACHIEVE SOUND ISOLATION AT WALLS (MAX : 2 PT)

For All Spaces:

Option 1: Noise isolation class

For walls that separate regularly occupied spaces the following requirements are met:

- The project meets the following minimum Noise Isolation Class (NIC) or Weighted Difference Level (Dw) for each wall type, as applicable. If an interior wall meets multiple categories listed, use the highest NIC/Dw value listed.

Interior Wall Type	Minimum NIC or Dw	OR	Minimum NIC or Dw for Projects That Achieve Feature S06.1
Between loud zones and regularly occupied spaces.	55	OR	N/A
Between areas for conferencing, learning or sleep and other regularly occupied spaces.	50	OR	45
Between adjacent quiet zones.	45	OR	40
Between rooms for concentration and other regularly occupied spaces.	40	OR	35
Between circulation zones and regularly occupied spaces.	35	OR	30

OR

Option 2: Speech privacy

For walls that separate regularly occupied spaces the following requirements are met:

- The sum of the measured Noise Isolation Class (NIC) or Weighted Difference Level (Dw) combined with the Noise Criteria Rating (NC) or A-weighted Sound Pressure Level (LAeq) within a room achieves the following minimum values, as applicable. If an interior wall meets multiple categories listed, use the highest value listed.

Source Room	Receiver Room	Minimum NIC + NC or Dw + LAeq
Enclosed loud zones	Open areas for concentration	80
	Circulation zones	
	All other occupiable areas	85
Enclosed areas for conferencing	Open areas for concentration	75
	Circulation zones	
	Enclosed quiet zones	80
Enclosed areas for learning	Enclosed areas for conferencing	85
Enclosed areas for sleeping	Enclosed areas for learning	
	Enclosed areas for sleeping	
Enclosed quiet zones	Open areas for concentration	70
	Circulation zones	
	Enclosed quiet zones	75
	Enclosed areas for conferencing	
	Enclosed areas for learning	80
	Enclosed areas for sleeping	

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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S04 REVERBERATION TIME | O (MAX : 2 PT)

Intent : Design spaces in accordance with comfortable reverberation times that support speech intelligibility, vocal effort and are conducive to concentration.

Summary : This WELL feature requires that steps be taken to address acoustical comfort, by controlling reverberation time based on room functionality.

Issue : Reverberation time is the length of time taken for a sound to decay by 60 dB from an initial impulse level. The length of reverberation time is a function of room size, surface area and sound absorbing properties of surface finishes. Spaces with longer reverberation times may be larger in volume with hard surfaces that reflect sound. Spaces with shorter reverberation may be smaller with softer surfaces that absorb sound. Spaces with high reverberation may have increased ambient noise levels and reduce speech intelligibility (e.g., public address, speech reinforcement and amplified speech). Studies have shown that high reverberation times in classrooms increase auditory workload in students and reduce cognition, memory retention and concentration.¹⁻⁴ Similarly, studies have shown that high reverberation reduces speech intelligibility among hard of hearing and non-native speaking populations.⁵⁻⁸

Solutions : Reverberation time can be controlled by adding absorptive surface finishes at ceilings, walls and furniture.⁹⁻¹¹ Projects that can alter room geometry can change layouts and room dimensions to support optimal reverberation times, as needed. Reducing reflective surfaces, such as glass, drywall, stone or similar, will also reduce reflected sound energy, which increases reverberation time. Reducing reverberation time also allows audio equipment for telecommunication/AV, speech reinforcement or public address to operate with higher speech intelligibility performance.^{5,12-14}

PART 1 ACHIEVE REVERBERATION TIME THRESHOLDS (MAX : 2 PT)

For All Spaces except Dwelling Units & Guest Rooms:

Option 1: Reverberation time, design

For projects in which the room types listed in the table cumulatively make up at least 10% of occupiable project area, the following requirements are met:

- a. Reverberation time is within the ranges shown in the following table:

Room Type	Room Volume, v (cubic meters)	Reverberation Time, t (seconds) ^{5,9,10}
	$v < 280 \text{ m}^3$	$t \leq 0.6$
Areas for learning	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 0.8$
Areas for lectures		
Areas for conferencing	$v > 570 \text{ m}^3$	$t \leq 1.0$
Areas with regularly used PA systems	N/A	$t \leq 1.5$
Areas for dining	N/A	$t \leq 1.0$
	$v < 280 \text{ m}^3$	$t \leq 0.8$
Areas for fitness	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 1.1$
	$v > 570 \text{ m}^3$	$t \leq 1.8$
	$v < 280 \text{ m}^3$	$t \leq 1.1$
Areas for music rehearsal	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 1.4$

Note : Where room types include multiple use types (e.g., learning and fitness) use the limits that include the lower reverberation time or range.

OR

Option 2: Reverberation time, performance

For projects in which the room types listed in the table cumulatively make up at least 10% of occupiable project area, the following requirements are met:

- a. Reverberation time is within the ranges shown in the following table:

Room Type	Room Volume, v (cubic meters)	Reverberation Time, t (seconds) ^{5,9,10}
Areas for learning	$v < 280 \text{ m}^3$	$t \leq 0.6$
Areas for lectures	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 0.8$
Areas for conferencing	$v > 570 \text{ m}^3$	$t \leq 1.0$
Areas with regularly used PA systems	N/A	$t \leq 1.5$
Areas for dining	N/A	$t \leq 1.0$
	$v < 280 \text{ m}^3$	$t \leq 0.8$

Areas for fitness	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 1.1$
	$v > 570 \text{ m}^3$	$t \leq 1.8$
Areas for music rehearsal	$v < 280 \text{ m}^3$	$t \leq 1.1$
	$280 \text{ m}^3 \leq v \leq 570 \text{ m}^3$	$t \leq 1.4$

Note : Where room types include multiple use types (e.g., learning and fitness) use the limits that include the lower reverberation time or range. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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S05 SOUND REDUCING SURFACES | O (MAX : 2 PT)

Intent : Design spaces with sound reducing surfaces to minimize the buildup of speech or other unwanted sound.

Summary : This WELL feature requires the use of acoustic materials that absorb and/or block sound to support concentration and reduce reverberation.

Issue : Increases in reverberation are commonly caused by lack of acoustical absorption.¹ When spaces are designed with hard reflective ceilings and walls, perception of noise increases with an increase in ambient, reflected noise that decays in level at a longer rate.^{2,3} When open collaborative spaces lack acoustical absorption at ceilings or partial height barriers, workers can become distracted by reflected sound, especially if that speech is perceptible.³⁻⁶ Conversely, in learning environments, reverberation can impair a listener's comprehension of speech due to poor speech intelligibility, especially when listeners are hard of hearing, expending a greater auditory workload to process speech resulting in reduced task performance.⁷⁻¹² In rooms where teleconferencing or audio equipment introduces amplified sound, spaces without adequate acoustical treatment are subject to higher functioning background noise levels, which may be heard in neighboring spaces.^{11,13-15} Lack of reflected sound control in spaces for conferencing can impact the quality of speech intelligibility over teleconferencing equipment.

Solutions : Providing acoustical treatment that significantly reduces sound across human speech frequencies can greatly reduce unwanted buildup of reflected sound.¹ Ceilings typically provide the greatest area of coverage where highly absorptive materials can offer the best performance in reducing reverberation and controlling speech intelligibility and occupied noise levels.^{3,16,17} Wall treatment can reduce reflections that impact performance of speech intelligibility with teleconferencing equipment.¹

PART 1 IMPLEMENT SOUND REDUCING SURFACES (MAX : 2 PT)

For All Spaces except Dwelling Units:

For projects in which the room types listed in the table cumulatively make up at least 10% of occupiable project area, the following requirements are met:

a. Acoustical furnishings meet the criteria shown in the following table:

Room Type	Metric	Tier 1 1 Point	Tier 2 2 Points
Open workspaces	Minimum NRC OR Alpha-w	0.75 for at least 75% of available ceiling area	0.90 for all available ceiling area ^{1,18}
	Minimum furniture height and NRC OR Alpha-w	N/A	Partial height barriers with a minimum height of 1.2 m above finished floor and a minimum NRC OR Alpha-w value of 0.70 between all opposing workstations ¹
Areas for conferencing and learning	Minimum NRC OR Alpha-w at ceilings	0.75 for at least 50% of available ceiling area	0.90 for all available ceiling area
	Minimum NRC OR Alpha-w at walls	0.75 on at least 25% of one wall	0.80 on at least 25% of two perpendicular walls
Areas for dining	Minimum NRC OR Alpha-w at ceilings	0.75 on at least 50% of available ceiling area	0.90 for all available ceiling area

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S06 MINIMUM BACKGROUND SOUND | O (MAX : 2 PT)

Intent : Increase acoustical privacy within and between occupied spaces.

Summary : This WELL feature requires the use of dedicated artificial sound to uniformly increase speech privacy between occupied spaces.

Issue : Speech privacy can be described as the signal-to-noise ratio, or sound blocking and sound cover, of perceptible speech reaching a listener.¹ Ambient background sound, which can include artificial sound sources, can increase speech privacy to comfortable levels.^{2,3} However, ambient background noise provided by HVAC systems alone is shown to fluctuate by zone and by as much as 15 dBA or more in a given year.⁴ Some HVAC systems, like chilled beams, offer little to no ambient sound contribution and result in quiet interiors, where speech is readily intelligible and distracting. In offices and patient rooms where speech privacy is critical between regularly occupied spaces, a combination of reliable background sound is needed where partitions do not perform above a minimum STC-30.⁵

Solutions : Consistent background sound can be delivered to a space using an adjustable array of loudspeakers located such that sound is uniformly distributed.^{6,7} These systems are programmed to output a sound source similar to the sound of air flowing through HVAC. The sound source is broad-band and designed to elevate ambient background sound level to improve the signal-to-noise ratio and reduce speech perception between enclosed and open environments.^{8,9}

PART 1 PROVIDE MINIMUM BACKGROUND SOUND (MAX : 1 PT)

For Office Spaces:

The following requirements are met:

- a. A sound masking system is installed in open areas and enclosed rooms designated as quiet zones, circulation zones and in areas where workstations are present.
- b. The sound masking system produces a 1/3 octave band adjustable output signal and minimum frequency spectrum of 100 Hz to 5 kHz.
- c. The sound masking system is commissioned such that the following sound pressure levels are not exceeded:
 1. Open areas designated as quiet zones, circulation zones and areas where workstations are present: 48 dBA.
 2. Enclosed rooms labeled as quiet zones: 42 dBA.
- d. The sound masking system is verified by a professional sound masking system installer in accordance with ASTM 1573-18 or equivalent standard.

PART 2 PROVIDE ENHANCED SPEECH REDUCTION (MAX : 1 PT)

For All Spaces:

The project achieves the following features and parts:

- a. Achieve two points in either Feature S03 Part 2 - Achieve Sound Isolation at Walls or Feature S05 Part 1 – Implement Sound Reducing Surfaces.
- b. Feature S06 Part 1 – Provide Minimum Background Sound.

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S07 B IMPACT NOISE MANAGEMENT | O (MAX : 3 PT)

Intent : Reduce the level of impact noise radiation by designing resilient floors.

Summary : This WELL feature requires projects to manage background noise levels by demonstrating compliance with impact noise mitigation techniques.

Issue : Sound can transmit between rooms within a building as structure-borne impact noise. This impact noise travels through structures (e.g., walls, floors, columns, piping) as vibrations that are then radiated as airborne noise to a listener.¹ Impact noise typically derives from objects impacting a floor (e.g., footfall, machinery, gym equipment) and can result in workplace distractions, sleep disturbance or disrupted focus.^{2,3}

Solutions : The overall construction of a building influences impact noise radiation levels. For example, a building that utilizes a light-weight floor construction (e.g., wood truss, cross-laminated timber, steel frame) generally exhibits higher degrees of impact noise radiation between floors.⁴⁻⁷ Conversely, buildings constructed with resilient, composite floor-ceiling construction (e.g., thick concrete slab, suspended ceiling, floor with an underlayment) generally exhibit lower degrees of impact noise radiation. The performance of floor-ceiling materials can be measured using the following metrics: Impact Insulation Class Rating (IIC), Normalized Impact Sound Rating (NISR) or Weighted Standardized Impact Sound Pressure Level (L_{nTw}).

PART 1 SPECIFY IMPACT NOISE REDUCING FLOORING (MAX : 1 PT)

For All Spaces:

The following requirements are met:

- For the following room types within the project boundary the floor-ceiling construction achieves the following minimum Impact Insulation Class (IIC) ratings with materials tested in accordance with ASTM E492-09, ISO 717.2 or predicted results determined through use of sound transmission modeling software (L_{nTw} may be used as an equivalent metric and values may be determined by subtracting the IIC values listed below from 110):

Room Type	Location of Applicable Floor-Ceiling Assembly	Minimum Impact Insulation Class (IIC) ¹⁰³
Quiet zones (except areas for concentration)	Above	55
Areas for fitness	Below	50
Enclosed areas for concentration and conferencing	Above	50
Open areas for concentration	Above	45
Areas for retail and dining	Below	45

Note :

This requirement does not apply to floor/ceiling assemblies that separate a relevant room type from a parking garage or non-occupiable space (e.g., a quiet zone that is vertically adjacent to a roof, equipment room or attic).

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 2 MEET THRESHOLDS FOR IMPACT NOISE RATING (MAX : 2 PT)

For All Spaces:

The following requirements are met:

- For the following room types within the project boundary the floor-ceiling construction achieves the following Normalized Impact Sound Ratings (NISR), as measured by a professional in acoustics, in accordance with ASTM E1007-19, ISO 16283 or equivalent (L_{nTw} may be used as an equivalent metric and may be determined by subtracting the NISR values listed below from 110):

Room Type	Location of Applicable Floor-Ceiling Assembly	Tier 1 1 point Minimum NISR ¹	Tier 2 2 points Minimum NISR ¹
Quiet zones (except areas for concentration)	Above	52	57
Areas for Fitness (If space is within the project boundary)	Below	47	52
Enclosed Areas for Concentration and Conferencing	Above	47	52

Open Areas for Concentration	Above	42	47
Areas for Retail and Dining (If space is within the project boundary)	Below	42	47

Note :

This requirement does not apply to floor/ceiling assemblies that separate a relevant room type from a non-occupiable space (e.g., a quiet zone that is vertically adjacent to a roof, equipment room or attic).

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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S08 ENHANCED AUDIO DEVICES | O (MAX : 2 PT)

Intent : Improve speech intelligibility and accessibility by providing dedicated, high-performance audio technology.

Summary : This WELL feature requires projects to implement organizational policies and provide occupants with devices that support enhanced speech intelligibility and bolster hearing accessibility in spaces intended for telecommunicating, instruction and public address.

Issue : The ability for people to comprehend speech is a fundamental consideration of universal design. Reduced or low speech intelligibility can negatively impact occupant satisfaction and well-being, especially for non-native speakers, individuals with hearing loss or neurodiverse populations.¹⁻⁹ Audio equipment used for communication can further decrease comprehension of spoken word when installed and used incorrectly.^{1,5,10-13} Additionally, increased auditory workload can impact task performance, resulting in a higher risk of misunderstanding, operational errors and accidents.¹⁴ In educational settings, increased vocal effort by teachers to overcome poor intelligibility has been linked to vocal strain, decreased job performance, lower quality of life, higher rates of leave or absence and resignation.¹⁵⁻²⁰ Improving speech intelligibility can support classroom participation for deaf and hard-of-hearing students, which is linked to improved scores for quality of life, social contact with peers and mental health.²¹

Solutions : Implementing audio systems can improve speech intelligibility for end-users in various environments. These systems include teleconferencing equipment in offices, speech reinforcement systems in classrooms and public address systems.^{22,23} To provide the best possible outcomes for users, systems should be commissioned by a professional in audio engineering in order to meet diverse occupant needs and create more accessible spaces.²⁴⁻²⁶

PART 1 PROVIDE ENHANCED SPEECH INTELLIGIBILITY (MAX : 1 PT)

For All Spaces:

The project meets the following requirements, as applicable:

- a. All rooms that are intended for conferencing, distance learning, or hybrid collaboration contain a combination of microphones, speakers, cameras and supportive audio components (e.g., amplifiers, digital signal processors) that are commissioned by a professional audio engineer.
- b. All public address systems used on a daily basis meet the following:
 1. Commissioned by a professional audio engineer in accordance with NFPA 72 (Annex D), BS 5839 Part 8, ISO 7240 Parts 16 and 19 or equivalent.²⁷
 2. Achieves a minimum STI 0.50 or CIS 0.75 in at least 50% of regularly occupied acoustically distinguishable spaces (ADS) when measured in accordance with IEC 60268-16 or equivalent and as indicated in a commissioning report, acoustical model or similar.²⁸
- c. Speech reinforcement systems are installed in at least 80% of classrooms and all spaces for large presentations (e.g., lecture hall, conference center) and meet the following:
 1. Designed to meet audio distribution requirements in accordance with ANSI/ASA S12.60 Part 1.^{29,30}
 2. Commissioned by a professional acoustician or audio engineer in accordance with ANSI/INFOCOMM A102.01:2017 or equivalent.

PART 2 PRIORITIZE AUDIO DEVICES AND POLICIES (MAX : 1 PT)

For All Spaces:

The project or organization supports individual acoustical needs through policies that meet at least three of the following requirements:

- a. All employer-provided audio devices are managed internally by a qualified professional (e.g., IT professional, mobile device manager) and expectations for use are covered in the employee handbook and/or during onboarding of new staff.^{31,32}
- b. Eligible employees can request alternative working arrangements to accommodate their individual acoustic comfort needs (e.g., option to work remotely, different workstation location).^{33,34}
- c. Signage is used to indicate both the location and intended activities of the quiet and mixed zones. A minimum of one daily quiet hour is scheduled.³⁵
- d. Eligible employees and distance learners (as applicable) are provided speech-enhancing audio devices (e.g., headsets with active digital signal processing, or noise-cancellation) upon request and at no additional cost or subsidized at least 50%.³⁶

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S09 B HEARING HEALTH CONSERVATION | O (MAX : 1 PT)

Intent : Increase access to resources and structured programming for employees who are at risk of occupational hearing loss.

Summary : This WELL feature requires projects or organizations to implement policies and programs that support hearing health conservation.

Issue: The World Health Organization reports that approximately 466 million people worldwide have disabling hearing loss, of which 34 million are children.¹ Estimates suggest that by the year 2050 over 900 million people will have disabling hearing loss.¹ Typical environments where hazardous sound levels can occur include occupational settings where loud equipment is present (e.g., industrial spaces, airports, fitness studios), as well as settings where occupants regularly engage with personal listening devices (PLDs) that exceed safe listening levels (e.g., call centers, distance learning, recreational devices). Presbycusis, also known as age-related hearing loss, comprises most hearing loss cases.^{2–4} Studies have shown that individuals with hearing loss over the age of 65 are at higher risk for developing cognitive disorders like dementia and Alzheimer's disease.^{2,5,6} In all forms, hearing loss can lead to exclusion from communication and may impact social and emotional health by leading to feelings of loneliness, isolation and frustration, particularly among older populations.⁷ The annual economic impact of hearing loss is an estimated US \$750 billion, which accounts for costs related to healthcare (excluding the cost of hearing devices), educational support and loss of productivity and well-being.⁸

Solutions: Hearing health conservation programs that raise awareness about the risks of hearing loss and encourage individuals to use personal protective devices (e.g., earplugs, earmuffs) may help prevent noise induced hearing loss.^{8,10} Organization-wide programs that focus on the reduction of occupational noise exposure, provide access to regular hearing health screenings for vulnerable occupants and provide relevant staff with necessary training for use of loud equipment have been found to help limit disabling hearing impairments, reduce fatigue and in some cases improve morale and work efficiency.⁹ Companies that deploy these programs have reported reductions in medical expenses and worker compensation costs.^{11–13}

PART 1 IMPLEMENT A HEARING HEALTH CONSERVATION PROGRAM (MAX : 1 PT)

For All Spaces:

1: Hearing health conservation program

The project maintains a hearing health conservation program that meets the following requirements:

- a. Provides hearing protection that is selected, fitted and maintained for all occupants at no cost to the occupant.¹⁴
- b. Demonstrates compliance with OSHA Code of Federal Regulations Title 29 Chapter XVII Part 1910 Subpart G, European Council Directive 89/391/CEE or equivalent.¹⁵⁻¹⁷
- c. Audiogram tests are made available to employees working in loud zones, at no cost in a room that meets ANSI S3.1-2018 (or equivalent) requirements for background noise levels, using calibrated audiometers as per the schedule below:
 1. Annually for all employees.⁹
 2. Pre-employment or during onboarding for all new employees.⁹
 3. Prior to initial assignment in a hearing-hazardous zone as determined by the hearing conservation program supervisor (see below).⁹
 4. At the time of reassignment out of a hearing-hazardous work area or job, as determined by the hearing conservation program supervisor (see below).⁹
 5. At the conclusion of employment.⁹

2: Hearing health conservation supervisor

The project designates a qualified hearing health supervisor whose responsibilities include the following:

- a. Coordinates at least one hearing health training (e.g., workshop, seminar) per year that educates all employees on the following topics:
 1. Hearing loss and well-being.⁹
 2. Audiometric results and hearing threshold levels.⁹
 3. Noise exposure levels.⁹
 4. Correct use of hearing protection.⁹
- b. Manages the use and purchasing of hearing protection, audiometers and noise measuring equipment.⁹
- c. Schedules annual audiometric evaluations for employees.⁹
- d. Conducts an annual audit to determine that the hearing health conservation program adheres to OSHA, European Directive or equivalent regional regulations.^{9,15,17}
- e. Provides educational resources on hearing health to employees upon request.

Note : This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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